

 Academic Capstone Project Proposal

Yandex **Go** Armenia

Ride-Hailing Market Intelligence Platform

Automated Data Collection, Relational Warehousing & Executive BI Analytics

 Python 3.11

 Playwright Web Automation

 MySQL Relational Warehouse

 Power BI & DAX

 Star Schema ETL

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Executive Summary

PROJECT VISION



Market Challenge

Ride-hailing pricing and availability fluctuate dynamically across locations and time periods. Accessing granular, historical market trends requires automated real-time observation.



Data Solution

An automated end-to-end data pipeline harvesting publicly observable estimates every 15 minutes, standardizing raw feeds into an enterprise MySQL data warehouse.



Strategic Value

Delivers executive Power BI dashboards unlocking pricing patterns, peak demand surges, district heatmaps, and predictive pricing models for market intelligence.

Core Business Objectives

SCOPE & GOALS



1. Automated Scraping

Build resilient Python & Playwright scrapers capturing fare estimates, ETAs, and surge rates from 50+ locations every 15 mins.



2. MySQL Warehouse

Architect a scalable relational warehouse storing thousands of time-series observations with full data integrity.



3. Star Schema Design

Develop optimized dimensional models with explicit Fact and Dimension tables for high-performance BI queries.



4. Python ETL Pipeline

Implement Pandas transformations for data cleaning, geo-tagging, missing value handling, and scheduled database ingestion.



5. Power BI Dashboards

Design a comprehensive multi-tab executive dashboard featuring advanced DAX metrics, interactive filters, and drill-downs.



6. Market Intelligence

Extract actionable business insights on surge pricing, airport premiums, district demand, and predictive fare modeling.

Key Research Questions

ANALYTICAL FOCUS

 **Dynamic Pricing Patterns:** When are ride fares most expensive throughout the day and week, and what triggers surge pricing spikes?

 **Geographic & District Variance:** Which pickup districts and inter-city routes exhibit the highest average fare per kilometer?

 **Traffic vs. ETA Correlations:** How severely does peak traffic congestion impact Estimated Time of Arrival (ETA) across major urban hubs?

 **Ride Tier Value Comparison:** How do Economy, Comfort, Comfort+, Minivan, and Business tiers compare in price, ETA, and availability?

 **Predictive Price Modeling:** Can historical pricing and environmental variables be leveraged to accurately forecast future fare surges?

End-to-End Architecture

DATA PIPELINE



Data Engineering Specs

ETL & MODELING

</> Automation & Scraping

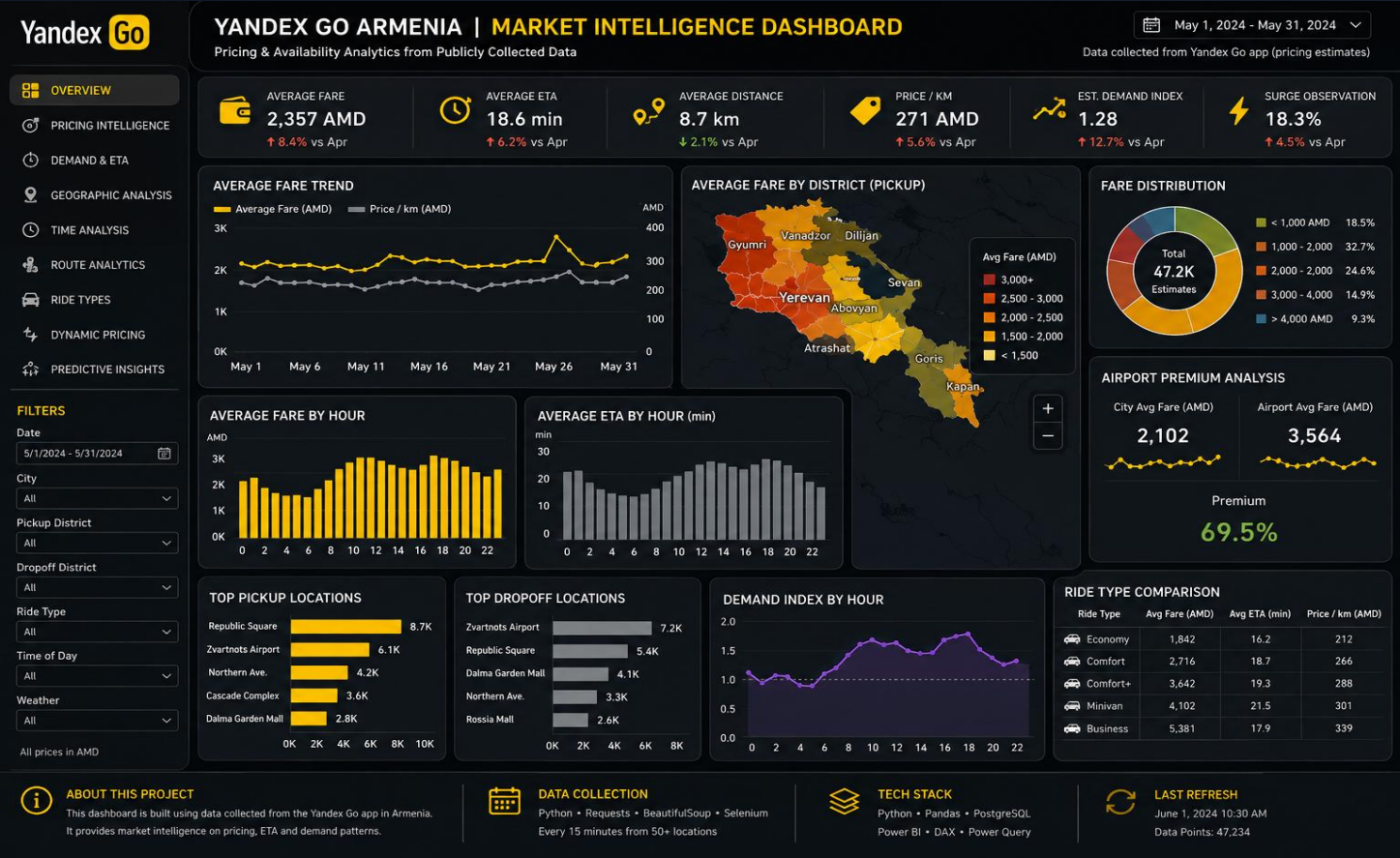
- **Frequency:** Cron-scheduled polling every 15 minutes across 50+ coordinate points in Armenia.
- **Technology:** Python 3.14 with Playwright async browser contexts for resilience.
- **Data Fields:** Pickup/Dropoff coords, Fare (AMD), ETA (mins), Distance (km), Ride Tier, Surge Multiplier, and Timestamp.

🗄️ Star Schema Warehouse

- **Fact Table:** FactRideEstimates containing metric values (fare, distance, ETA, surge index).
- **Dimension Tables:** DimLocation, DimTime, DimDate, DimRideType, and DimWeather.
- **Optimization:** Indexing on timestamp and location keys for sub-second DAX query execution.

Power BI Dashboard & Report

8 INTERACTIVE VIEWS



Implementation Roadmap

PROJECT TIMELINE

1

Phase 1: Scraping

- Setup Playwright scripts
- Define location matrix
- JSON parser development
- 15-min automated cron

2

Phase 2: Database

- MySQL schema setup
- Star Schema creation
- Index optimization
- Historical data backfill

3

Phase 3: Business Intelligence

- Power BI integration
- DAX measures setup
- Dashboard views
- Interactivity & filters

4

Phase 4: ML & Delivery

- Predictive fare models
- Executive report
- Technical documentation
- GitHub repo release

Expected Project Scope

DELIVERABLES

Deliverable Component	Technical Stack	Key Output / Description
 Automated Data Scraper	Python + Playwright	Headless automated browser pipeline extracting JSON pricing every 15 min.
 Relational Data Warehouse	MySQL Engine	Star schema relational store with FactRideEstimates & 5 dimension tables.
 Data Transformation Pipeline	Python + Pandas	Automated ETL script handling data validation, cleaning, and database loading.
 Power BI Executive Suite	Power BI Desktop + DAX	Interactive 8-tab dashboard deck with real-time dynamic filters and heatmaps.
 Project Portfolio & Docs	GitHub Repository	Complete source code, technical documentation, SQL scripts, and project writeup.

Questions & Discussion

Thank you for reviewing this project proposal!



GitHub Project Repository

github.com/edita-avetisyan

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